

CYBER SECURITY



Module I: Introduction & Cyber Security Fundamentals

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Lecture Agenda

- Overview of the module and assessment structure
- Introduction to cyber security and its importance
- Understanding the threat landscape and attackers
- CIA triad explained with practical examples
- Introduction to cryptography and its role
- In-class activities and preparation for next week

Module Overview

- This module introduces the foundations of cyber security and applied cryptography
- Students learn how digital systems are protected against modern cyber threats
- The module combines theoretical knowledge with practical security concepts

Why Cyber Security Is Important

- Cyber attacks are increasing in frequency and sophistication worldwide
- Security breaches can cause financial loss and reputational damage
- Cyber security is essential for governments, organisations, and individuals

Definition of Cyber Security

- Cyber security refers to the protection of systems, networks, and digital data
- It prevents unauthorised access, misuse, and disruption of information
- Cyber security applies across hardware, software, and human processes

Core Security Objectives

- Confidentiality ensures data is accessible only to authorised users
- Integrity ensures data is accurate and has not been altered
- Availability ensures systems and data are accessible when required

CIA Triad Explained

- Confidentiality is achieved using encryption and access controls
- Integrity is protected using hashing and validation mechanisms
- Availability is maintained using backups, redundancy, and fault tolerance

Cyber Threat Landscape

- Malware attacks include viruses, worms, and ransomware
- Phishing attacks exploit human trust to steal information
- Insider threats arise from employees or contractors

Common Cyber Attacks

- Denial-of-Service attacks disrupt access to systems
- Man-in-the-Middle attacks intercept communications
- Ransomware encrypts data and demands payment

Threat Actors

- Cyber criminals are motivated by financial gain
- Hacktivists are driven by political or social causes
- Nation states conduct cyber espionage and warfare

Introduction to Cryptography

- Cryptography uses mathematical techniques to secure information
- It supports confidentiality, integrity, and authentication
- Cryptography enables secure communication over insecure networks

Why Cryptography Matters

- Protects sensitive personal and organisational data
- Builds trust in online services and digital transactions
- Forms the foundation of modern cyber security systems

Ethical and Legal Responsibilities

- Cyber security professionals must act ethically and responsibly
- Unauthorised system access is illegal
- Legal compliance and data protection are mandatory

Preparation for Next Week

- Next topic: Security Frameworks